

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Summative Assessment 1

Summative Assessment (SA-1) - Answer Key

Class: Class 10

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Networking Concepts, Internet and Web Technologies, HTML
Fundamentals, CSS Styling

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: MCQ (10 x 1 = 10 marks)

1. **Answer:** d - Transport layer
 - **Explanation:** The Transport layer (Layer 4) handles end-to-end communication, reliability, and flow control using TCP/UDP.
2. **Answer:** c - 192.168.1.1
 - **Explanation:** Class C addresses range from 192.0.0.0 to 223.255.255.255. 192.168.1.1 is a Class C private address.
3. **Answer:** c - Tells the browser this is an HTML5 document
 - **Explanation:** `<!DOCTYPE html>` declares the document type as HTML5, ensuring the browser renders in standards mode.
4. **Answer:** b - padding
 - **Explanation:** Padding creates space between the element's content and its border. Margin is outside the border.
5. **Answer:** c - `<article>`
 - **Explanation:** `<article>` represents a self-contained composition like a blog post or news article.
6. **Answer:** b - Defines character encoding
 - **Explanation:** `meta charset="UTF-8"` specifies the character encoding for the HTML document, supporting all languages.
7. **Answer:** c - 404
 - **Explanation:** HTTP 404 indicates "Not Found". 200 is OK, 301 is Moved Permanently, 500 is Internal Server Error.
8. **Answer:** c - ID selector `#main`
 - **Explanation:** ID selectors have highest specificity (1,0,0), followed by class (0,1,0), element (0,0,1), and universal (0,0,0).
9. **Answer:** a - `<video>`
 - **Explanation:** The `<video>` tag embeds video content in HTML5 with attributes like `src`, `controls`, `autoplay`.
10. **Answer:** b - Positions element relative to its normal position until scroll threshold
 - **Explanation:** `position: sticky` toggles between relative and fixed based on scroll position.

Part B: Short Answer (8 x 2 = 16 marks)

1. **IPv4 vs IPv6**
 - **Answer:** IPv4: 32-bit address, 4 billion addresses, dotted decimal notation (192.168.1.1), no built-in security. IPv6: 128-bit address, unlimited addresses, hexadecimal notation

(2001:0db8::1), built-in IPSec security. IPv6 was introduced due to IPv4 address exhaustion and need for better security.

- **Marking:** 1 mark for comparison, 1 mark for why introduced.

2. HTML Form Code

- **Answer:**

```
<form action="/submit" method="POST">
  <label for="name">Full Name:</label>
  <input type="text" id="name" name="name" required><br>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required><br>

  <label for="phone">Phone:</label>
  <input type="tel" id="phone" name="phone" pattern="[0-9]{10}" placeholder="10-digit number"><br>

  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob"><br>

  <label>Gender:</label>
  <input type="radio" id="male" name="gender" value="male">
  <label for="male">Male</label>
  <input type="radio" id="female" name="gender" value="female">
  <label for="female">Female</label><br>

  <input type="submit" value="Submit">
</form>
```

- **Marking:** 1 mark for form structure, 1 mark for validation attributes.

3. Block-level vs Inline elements

- **Answer:** Block-level: start new line, full width (div, p, h1-h6, ul, ol, li). Inline: no new line, content width only (span, a, img, strong, em). Inline-block: inline flow but accepts block styling (img, button, input).
- **Marking:** 1 mark for each type with examples.

4. CSS Box Model

- **Answer:** Content: actual content area. Padding: transparent space between content and border. Border: edge around padding. Margin: transparent space outside border. box-sizing: border-box includes padding and border in element's total width/height.
- **Marking:** 0.5 marks for each component, 0.5 marks for box-sizing.

5. TCP/IP Model

- **Answer:** Four layers: (1) Application - HTTP, FTP, SMTP, DNS; (2) Transport - TCP, UDP; (3) Internet - IP, ICMP, ARP; (4) Network Access - Ethernet, Wi-Fi. Each layer communicates with adjacent layers.
- **Marking:** 1 mark for layers, 0.5 marks for each protocol.

6. CSS Rules

- **Answer:**

```
/* Responsive nav */
nav { display: flex; flex-wrap: wrap; }
/* Hover underline */
a:hover { text-decoration: underline; }
```

```
/* Media query */
@media (max-width: 768px) { nav { flex-direction: column; } }
/* 3-column grid */
.grid { display: grid; grid-template-columns: repeat(3, 1fr); }
```

- **Marking:** 0.5 marks for each rule.

7. Responsive Web Design

- **Answer:** RWD creates websites that adapt to different screen sizes. Techniques: Media queries apply styles based on device width. Flexbox provides flexible layout. Relative units (% , em, vw) scale with viewport. Mobile-first approach designs for smallest screens first.
- **Marking:** 1 mark for definition, 1 mark for techniques.

8. Class vs ID selectors

- **Answer:** Class (.name): reusable on multiple elements, lower specificity. ID (#name): unique to one element, higher specificity. Specificity: ID > class > element. Example: #header { } overrides .header { } which overrides header { }.
- **Marking:** 1 mark for difference, 1 mark for specificity.

Part C: Long Answer (4 x 3 = 12 marks)

1. Student Management System HTML Page

- **Answer:** Complete page with: HTML5 boilerplate, semantic header with school name/logo, form with input validation (name, roll, class, section, marks), styled table with 5 rows, CSS with borders, colours, hover effects, responsive footer. Key marks: proper structure, working form, styled table, responsive design.
- **Marking:** 1 mark for structure, 1 mark for form, 1 mark for table/CSS.

2. Bus, Star, and Mesh Topologies

- **Answer:**
- **Bus:** Linear backbone cable, all devices connect to it. Data travels both directions. Advantages: simple, cheap. Disadvantages: single point of failure, hard to troubleshoot. Best for small networks.
- **Star:** Central hub, all devices connect to it. Data passes through hub. Advantages: easy to install, fault isolation. Disadvantages: hub failure, more cable. Best for offices.
- **Mesh:** Every device connects to every other. Data takes multiple paths. Advantages: redundant, reliable. Disadvantages: expensive, complex. Best for critical systems.
- **Marking:** 1 mark for each topology (structure, flow, pros/cons).

3. Restaurant Menu Webpage

- **Answer:** Complete CSS-styled page with: hero section with gradient background, menu section using CSS Grid, food items in categories, responsive design with media queries, typography with Google Fonts, colour scheme matching restaurant theme.
- **Marking:** 1 mark for hero, 1 mark for menu layout, 1 mark for responsiveness/styling.

4. DNS Resolution Process

- **Answer:** DNS translates domain names to IP addresses. Types: Recursive (queries on behalf of client), Root (directs to TLD servers), TLD (manages .com, .org etc.), Authoritative (final answer). Process: Browser checks cache → OS cache → Recursive resolver → Root server → TLD server → Authoritative server → Returns IP to browser.
- **Marking:** 1 mark for DNS purpose, 1 mark for server types, 1 mark for process.

Part D: Application Based (2 x 1 = 2 marks)

1. Responsive Navigation Fix

- **Answer:** Problem: Nav disappears entirely on mobile. Fix: Use hamburger menu pattern.

```
nav { display: flex; flex-wrap: wrap; }
.nav-links { display: flex; list-style: none; }
@media (max-width: 768px) {
  .nav-links { display: none; }
  .menu-toggle { display: block; }
  .nav-links.active { display: flex; flex-direction: column; }
}
```

- **Marking:** 1 mark for identifying problem, 1 mark for solution.

2. Office Network Recommendation

- **Answer:** Use WAN (Wide Area Network) to connect offices across cities. Protocols: TCP/IP for reliable data transfer, VPN for secure communication. Routers at each office connect to ISP. Video conferencing uses RTP/RTCP protocols.
- **Marking:** 0.5 marks for network type, 0.5 marks for protocols.